Exercise semester 1/2569

Exercise 2 Schedule management (critical path method) (10 points)

Submit your answer in a **pdf file**. Name the file by **your name and student ID**.

The table below shows the network diagram data of a project. All estimated durations are in days.

Activity	Predecessor	Estimated duration
A	-	2
B	A	2
C	A	3
D	A	4
E	B	2
F	C	3
G	D	6
H	E, F	2
I	E, F	5
J	G, I	1
K	H, J	2

1. Draw a PDM (activity on node) network diagram of the project and calculate the ES, EF, LS, LF, TS, FS of each activity (similar to slide 33 of chapter 6).

2. Identify all paths on the network diagram and their lengths.

3. The critical path for this project is

It takes ..... days.

4. The shortest possible time needed to complete this project is ..... days.

5. Which activities may be delayed and their delay will not affect the immediately following activities? Explain.

6. Which activities may be delayed and their delay will not affect the project finish date? Explain.

7. If B is delayed and starts on day 5, choose one.

☐ The project finish date is delayed for at least ..... day(s).

☐ The project finish date is not affected by this delay.

Explain your choice.

8. If H is delayed and starts on day 10, choose one.

☐ The project finish date is delayed for at least ..... day(s).

☐ The project finish date is not affected by this delay.

Explain your choice.

9. If D is delayed and starts on day 3, choose one.

☐ This delay affects G and J.

☐ This delay does not affect G and J.

☐ This delay affects G but does not affect J.

☐ This delay does not affect G but affects J.

Explain your choice.